

Probability and Statistics | Quiz-03
BSDSF23 (Afternoon)

Name: _____ **Roll Number:** _____

Instructor: Dr. Faisal Bukhari

Time allocated: 30 minutes

Total Marks: 50

Obtained Marks: _____

Q1: Consider the digits 0, 1, 2, 3, 4, 5, and 6. How many three-digit numbers can be formed using these digits if each digit can only be used once? [5]

Q2: In how many ways can 5 different trees be planted in a circle? [5]

Q3: A California study concluded that following 7 simple health rules can extend a man's life by 11 years on the average and a woman's life by 7 years. These 7 rules are as follows: no smoking, get regular exercise, use alcohol only in moderation, get 7 to 8 hours of sleep, maintain proper weight, eat breakfast, and do not eat between meals. In how many ways can a person adopt 5 of these rules to follow if the person never drinks and always eats breakfast?

[5]

Q4: According to a study published by a group of University of Massachusetts sociologists, approximately 60% of the Valium users in the state of Massachusetts first took Valium for psychological problems. Find the probability that among the next 8 users from this state who are interviewed

[7.5 + 7.5 = 15]

(a) exactly 3 began taking Valium for psychological problems

(b) at least 5 began taking Valium for problems that were not psychological

Q5: If 7 cards are dealt from an ordinary deck of 52 playing cards, what is the probability that **[10 + 10 = 20]**

(a) exactly 2 of them will be face cards?

(b) at least 1 of them will be a queen?